STM Exercise 5

Phase Diagrams

(full solubility in the solid state)

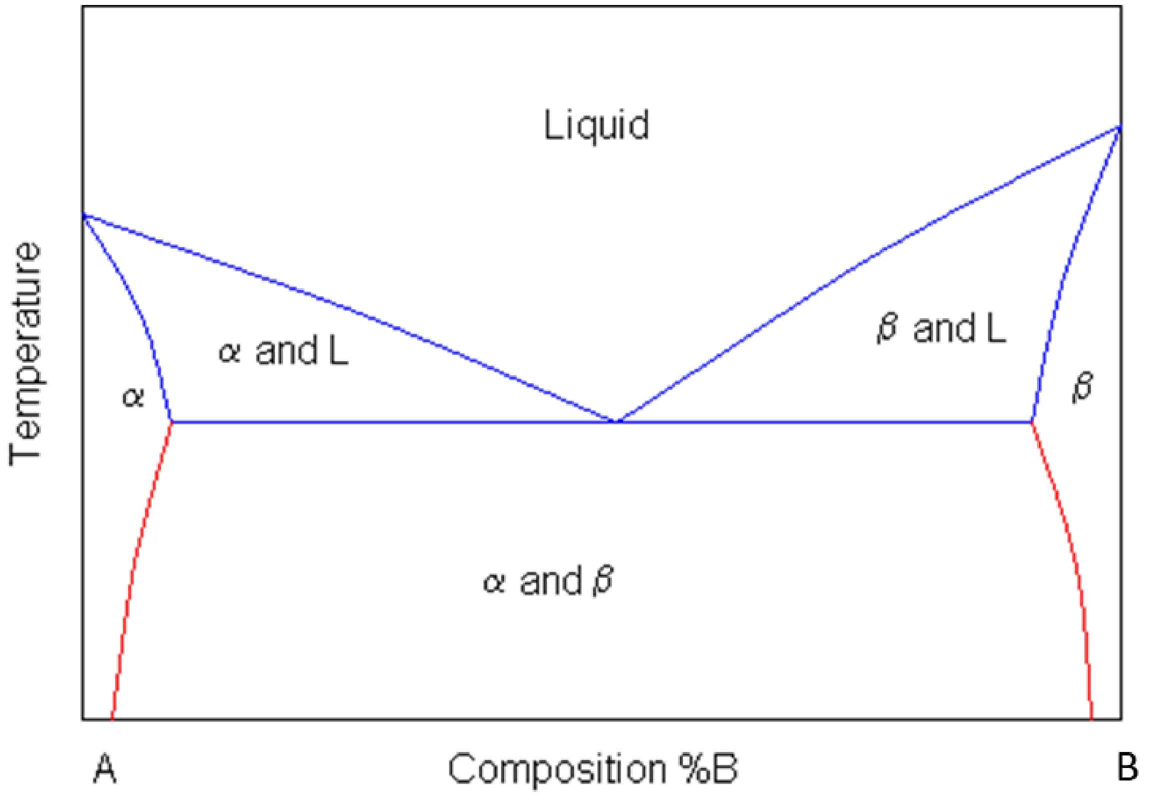
(no solubility in the solid state)

Phase Diagram

it shows the equilibrium states of a mixture, so that given a temperature and composition, it is possible to calculate which phases will be formed, and in what quantities.

Where **L** stands for liquid, and **A** and **B** are the two components and **α and β** are two solid phases rich in A and B respectively.

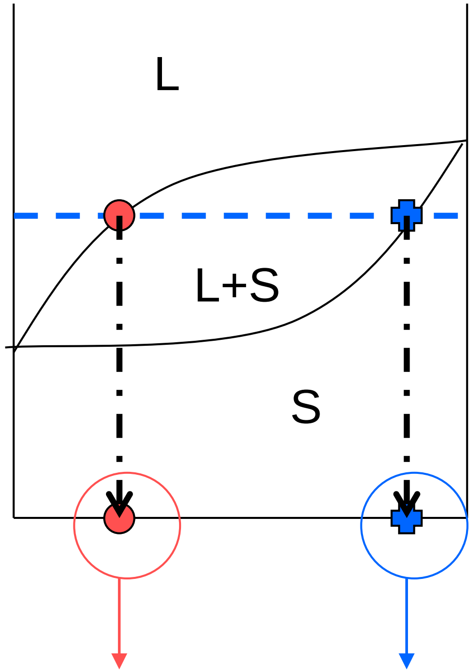
The blue lines represent the liquidus and solidus lines



The red lines involve a solid-to-solid transition (solvus lines)

Determination of the compositions of phases in equilibrium in a biphasic zone:

HORIZONTAL RULE



Liquid composition

Solid composition

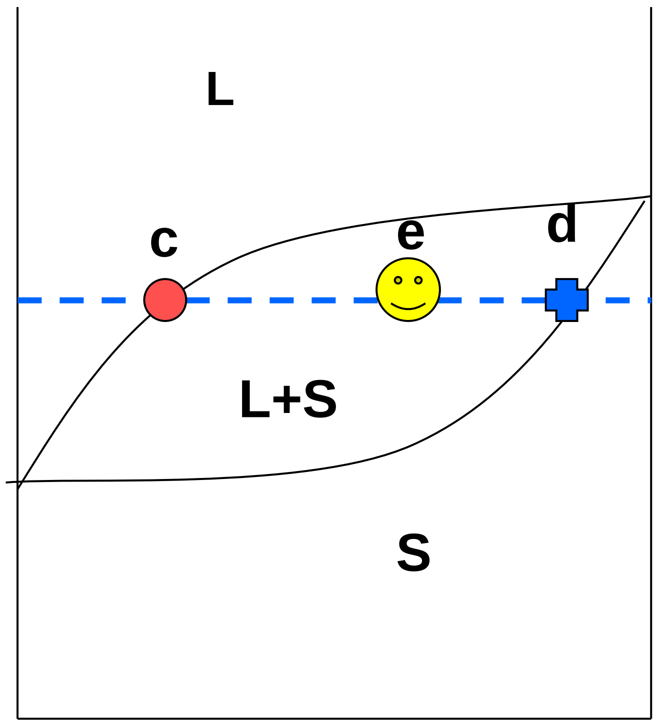
- Draw the tie line - - - - - -
- Individuate intersections with **liquidus** (•) and **solidus** ( ) lines
- The **composition of the liquid** is the value on the x axix correspondent to the intersection between the tie line and the liquidus line.
- The **composition of the solid** is the value on the x axis correspondent to the intersection of the tie line with the solidus line.

LIQUIDUS CURVE = **above** it **only liquid** is present

SOLIDUS CURVE = **below** it **only solid** is present

Phases relative amounts determination in a biphasic zone, at equilibrium:

LEVER RULE



- ➢Draw the tie line - - - - - -
- ➢Individuate the intersection between tie line and liquidus line (● c) and solidus line (**d**)
- ➢The % of phases in a point of the biphasic region (e) is defined as:

%L = (ed/cd)*100

%S = (ce/cd)*100

The lever rule

If an alloy consists of more than one phase, the amount of each phase present can be found by applying **the lever rule** to the phase diagram.

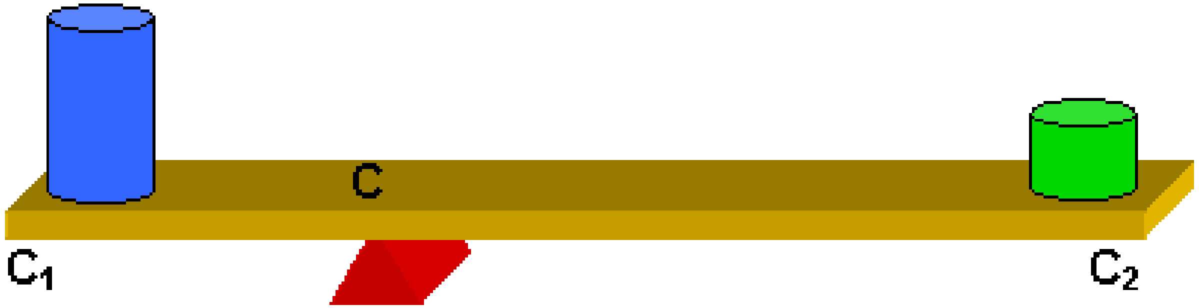
The lever rule can be explained by considering a simple balance.

The composition of the alloy is represented by the fulcrum C, and the compositions of the two phases by the ends of a bar (C_1 and C_2).

Each phase are determined by the weights needed to balance the system.

Fraction of phase 1 = $(C_2 - C) / (C_2 - C_1)$

Fraction of phase 2 = $(C - C_1) / (C_2 - C_1)$.



Exercise 1: Draw a phase diagram (for generic components A and B), with complete solubility in the liquid state and **complete solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

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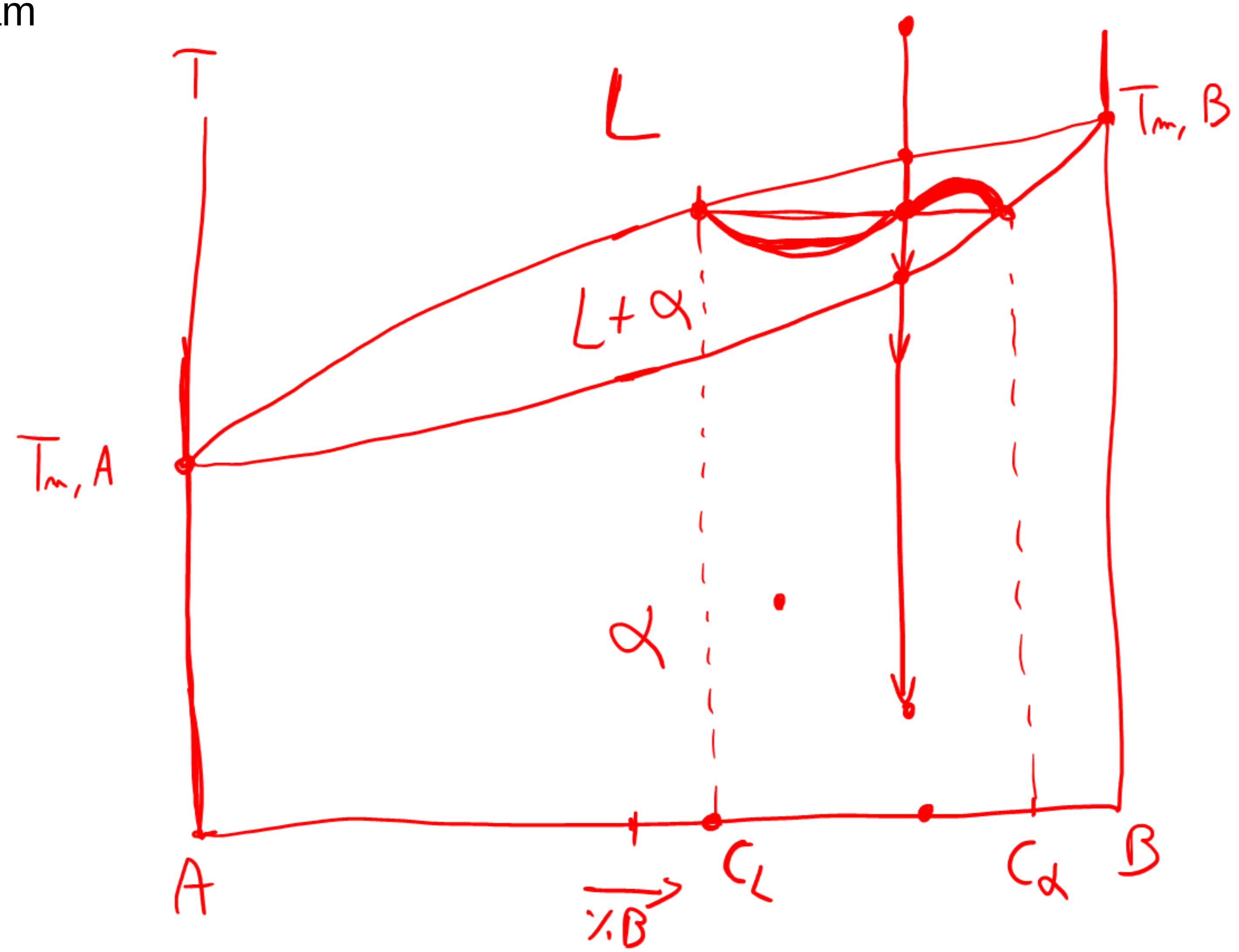
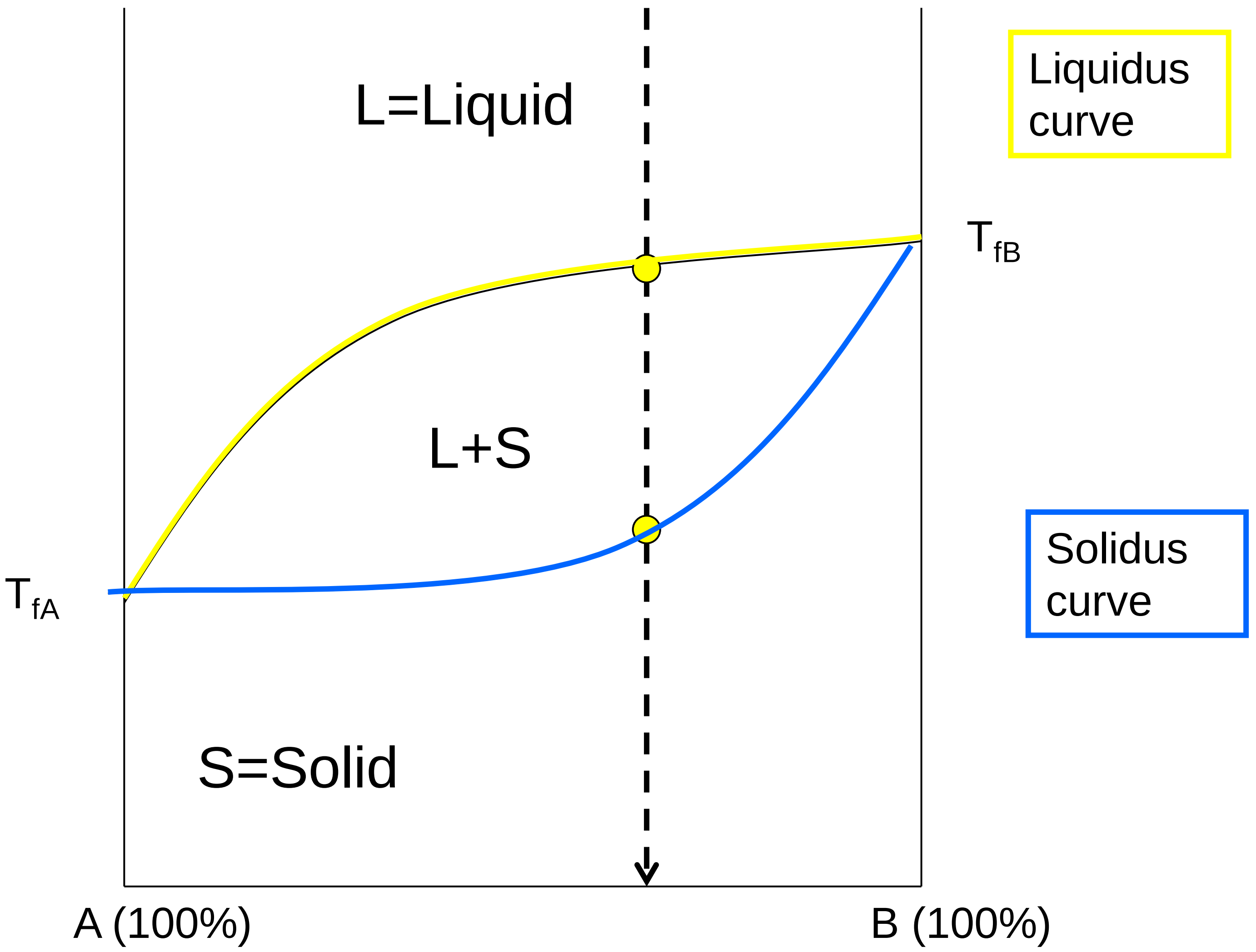
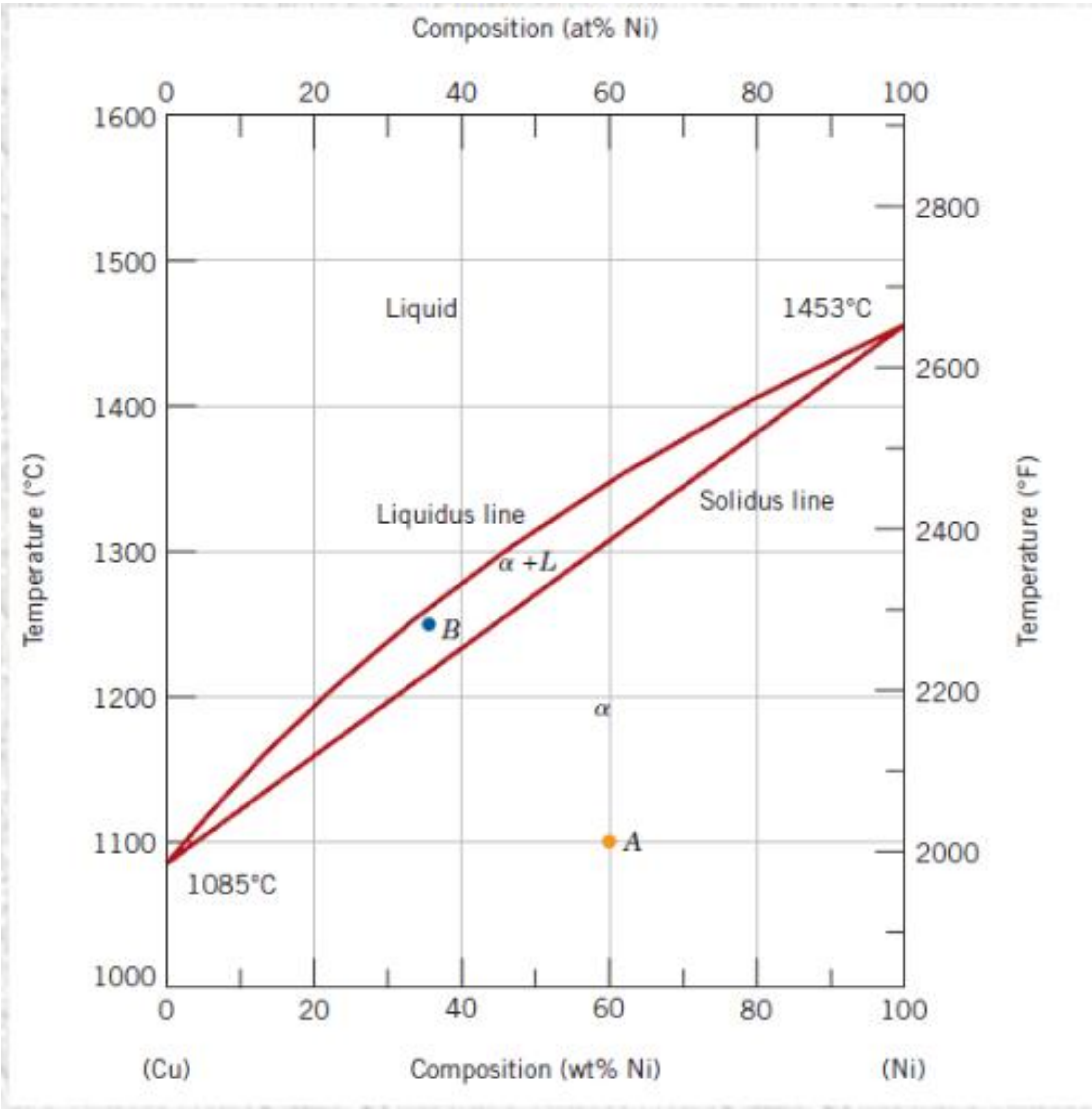


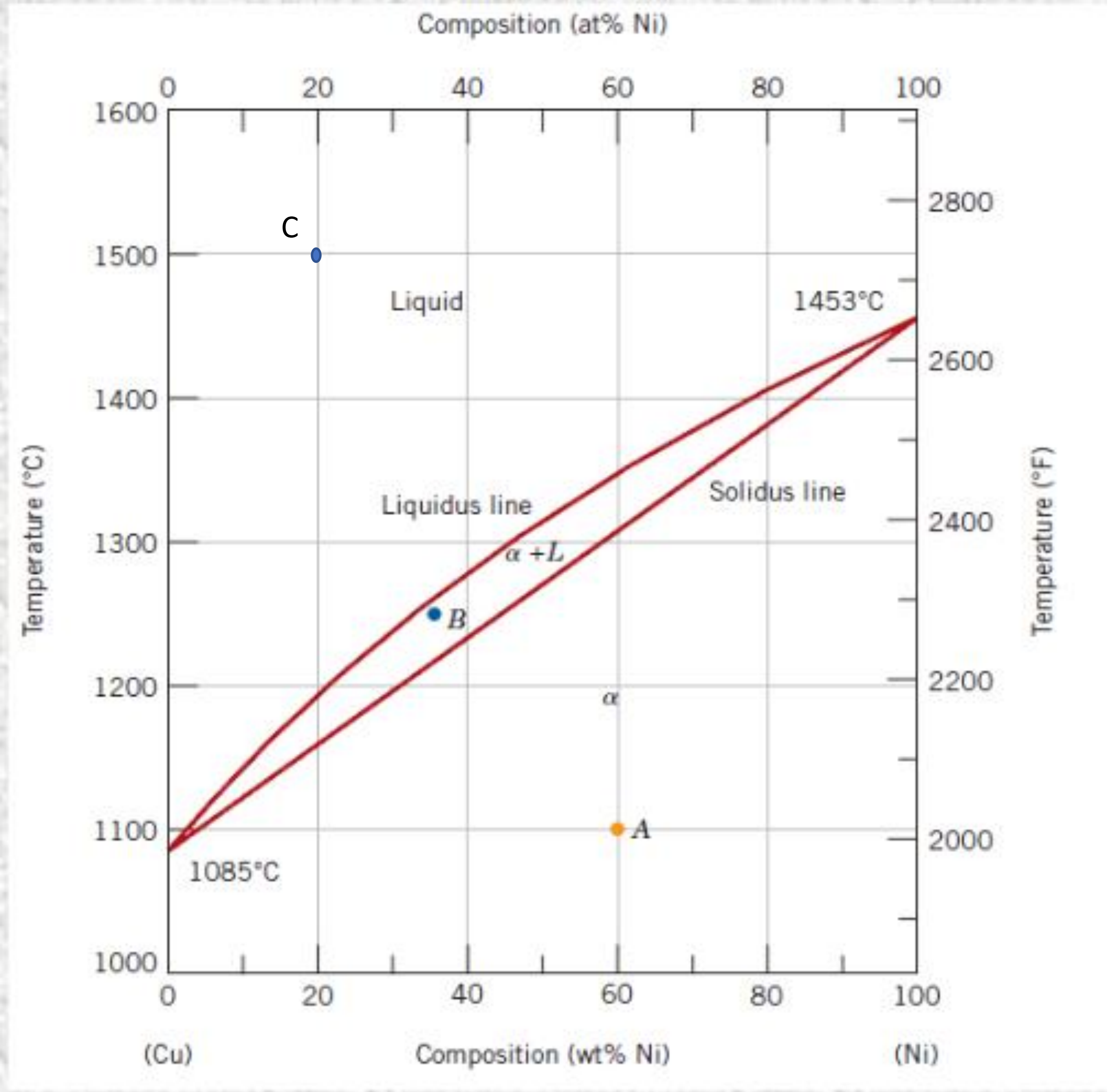
FULL SOLUBILITY IN THE SOLID AND LIQUID STATE

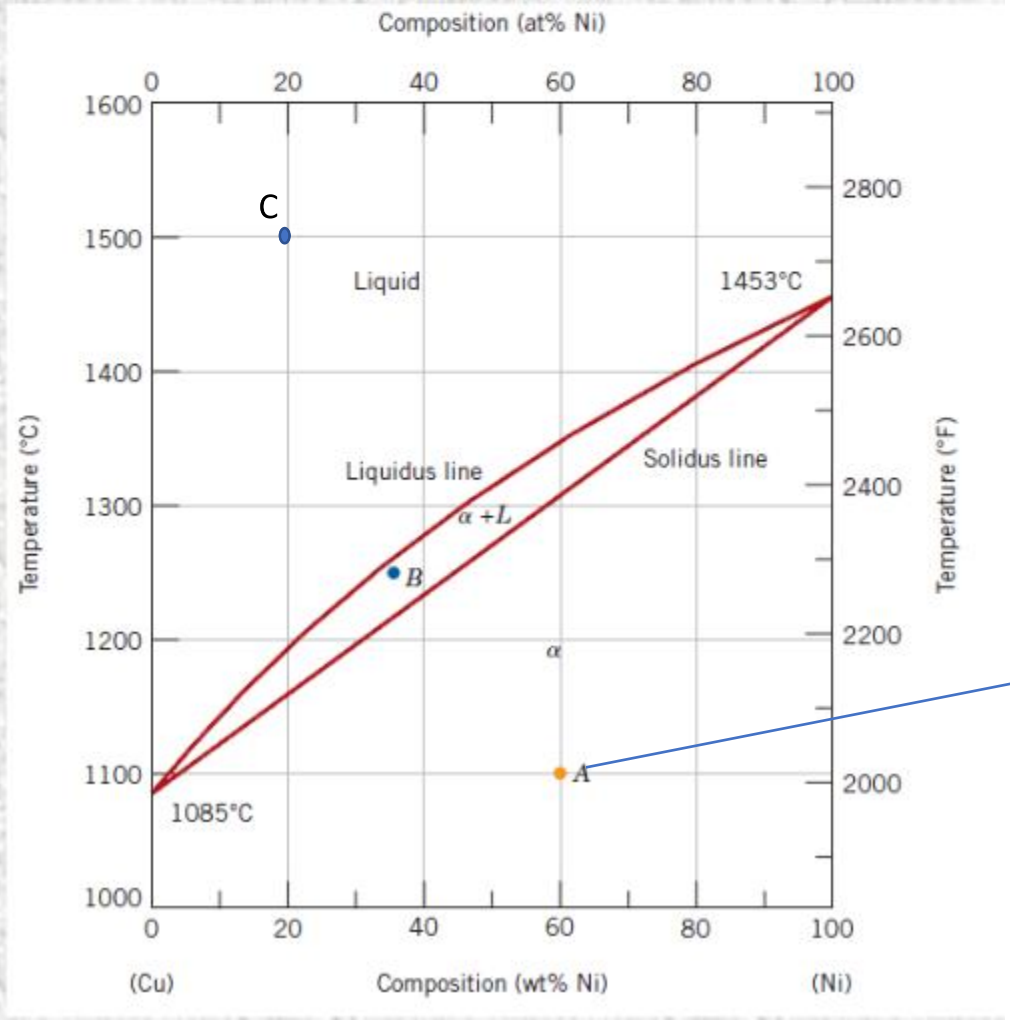


Exercise 1: full solubility in the solid state



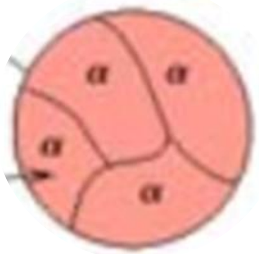
Exercise 2: determine the phases present, their composition and relative amounts in the point A, B and C indicated in the phase diagram. Schematize the microstructure in each point.

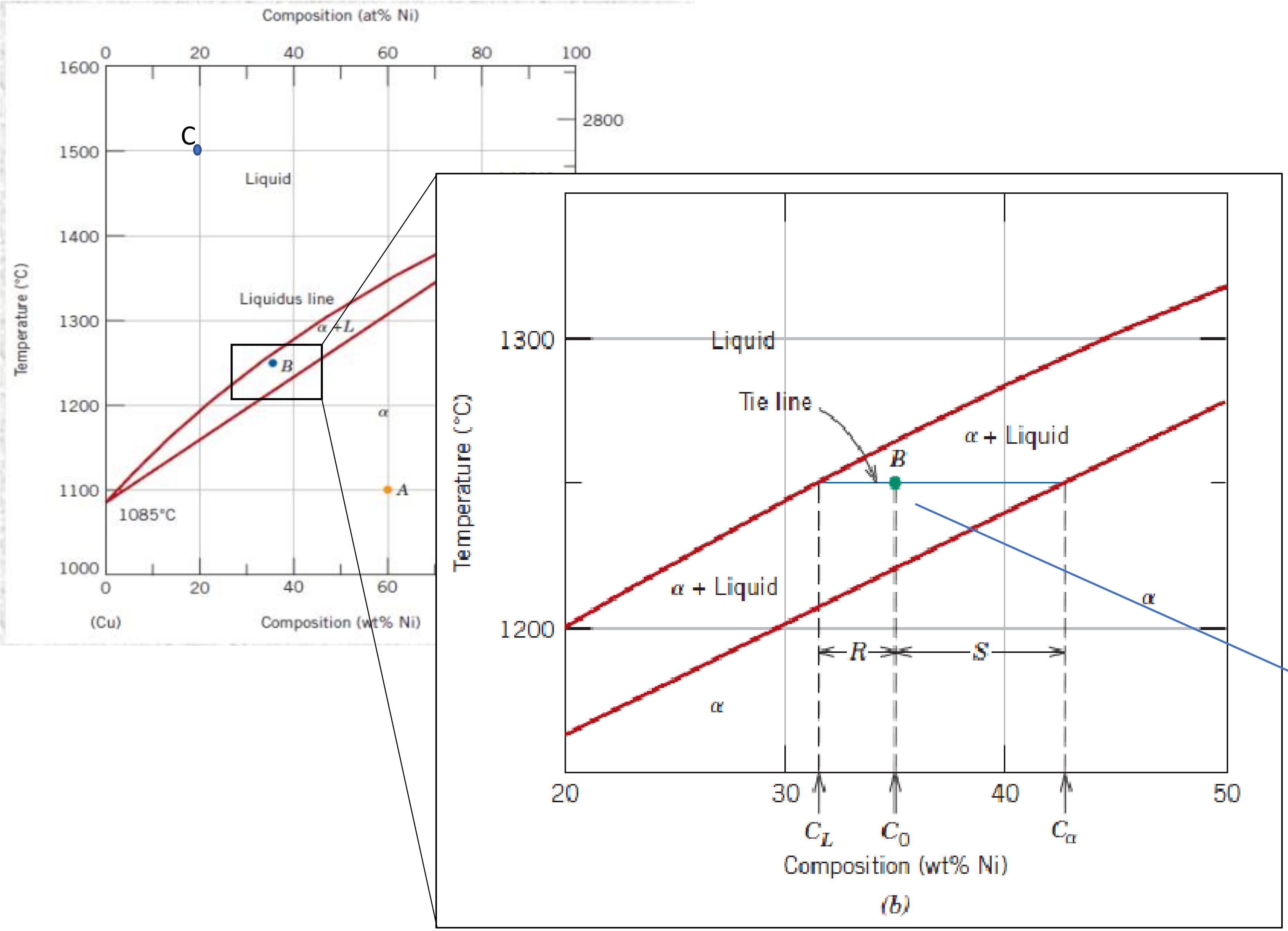




Point A:

Phases: 1 phase solid
Composition: 60%Ni, 40% Cu
Relative amount: 100% solid phase





<u>**Point B:**</u>

Phases: 2 pahses solid (α) +liquid (L)

Composition: 2 phases region → <u>horizontal rule</u>

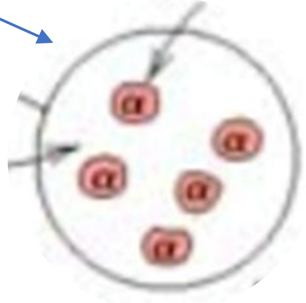
Composition of liquid (L): CL= 32% Ni, 68%Cu

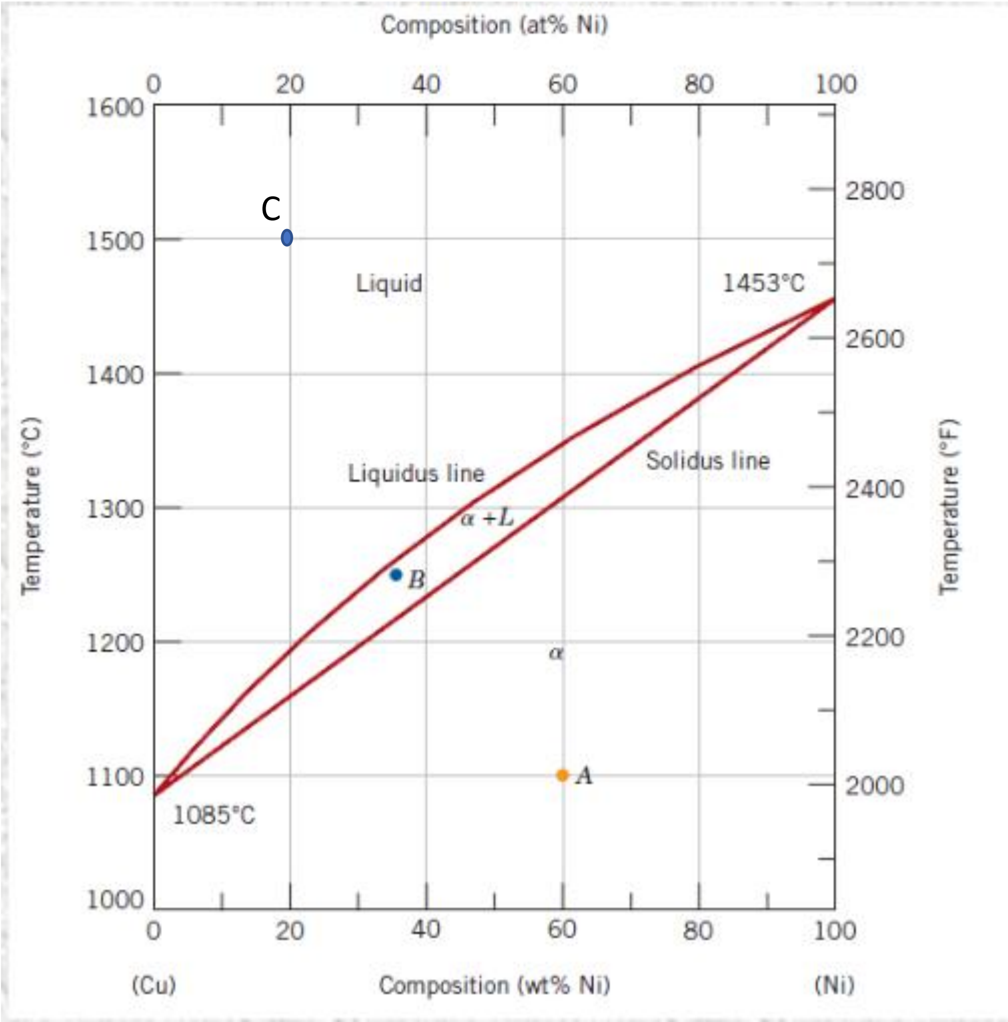
Composition of solid (α): C α= 43%Ni, 57% Cu

Relative amount: 2 phases region → <u>lever rule</u>

% α=R/(R+S)*100= (35-32)/(43-32)*100=27%

%L=S/(R+S)*100=(43-35)/(43-32)*100=73%

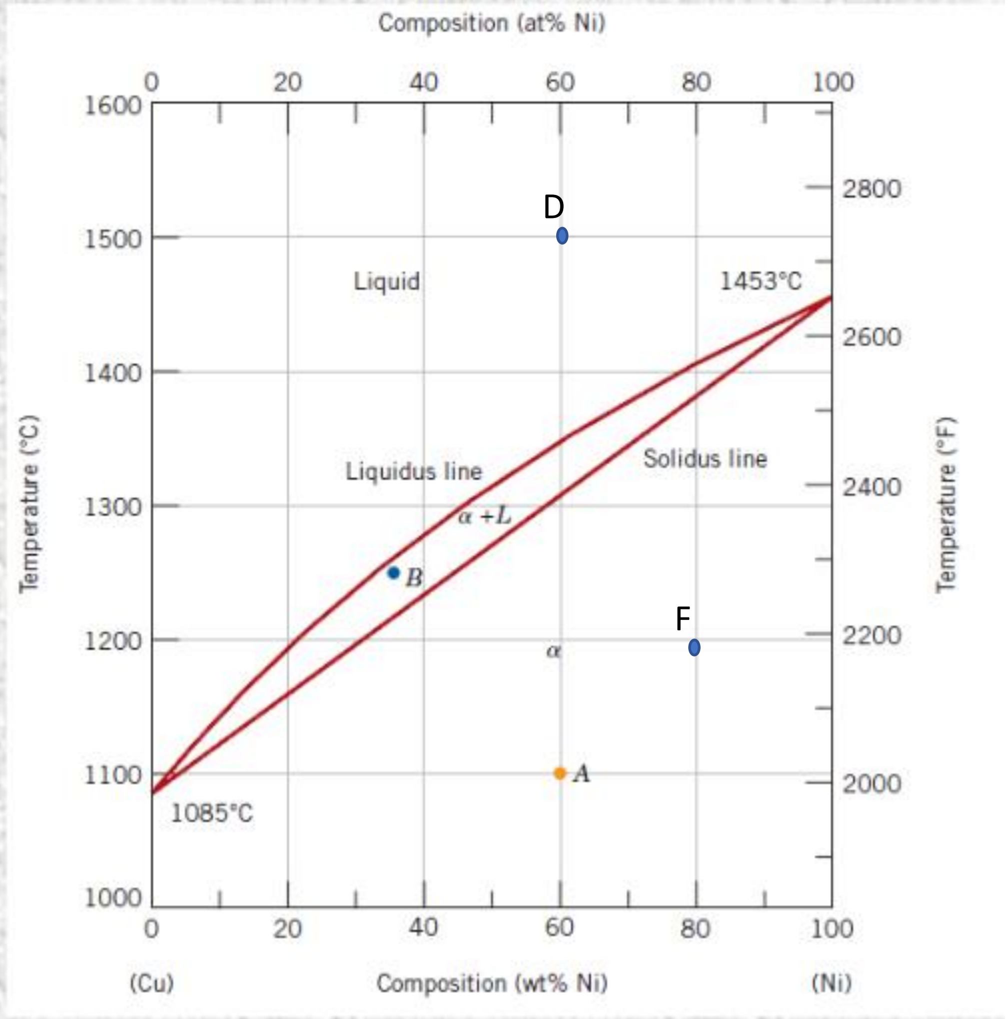


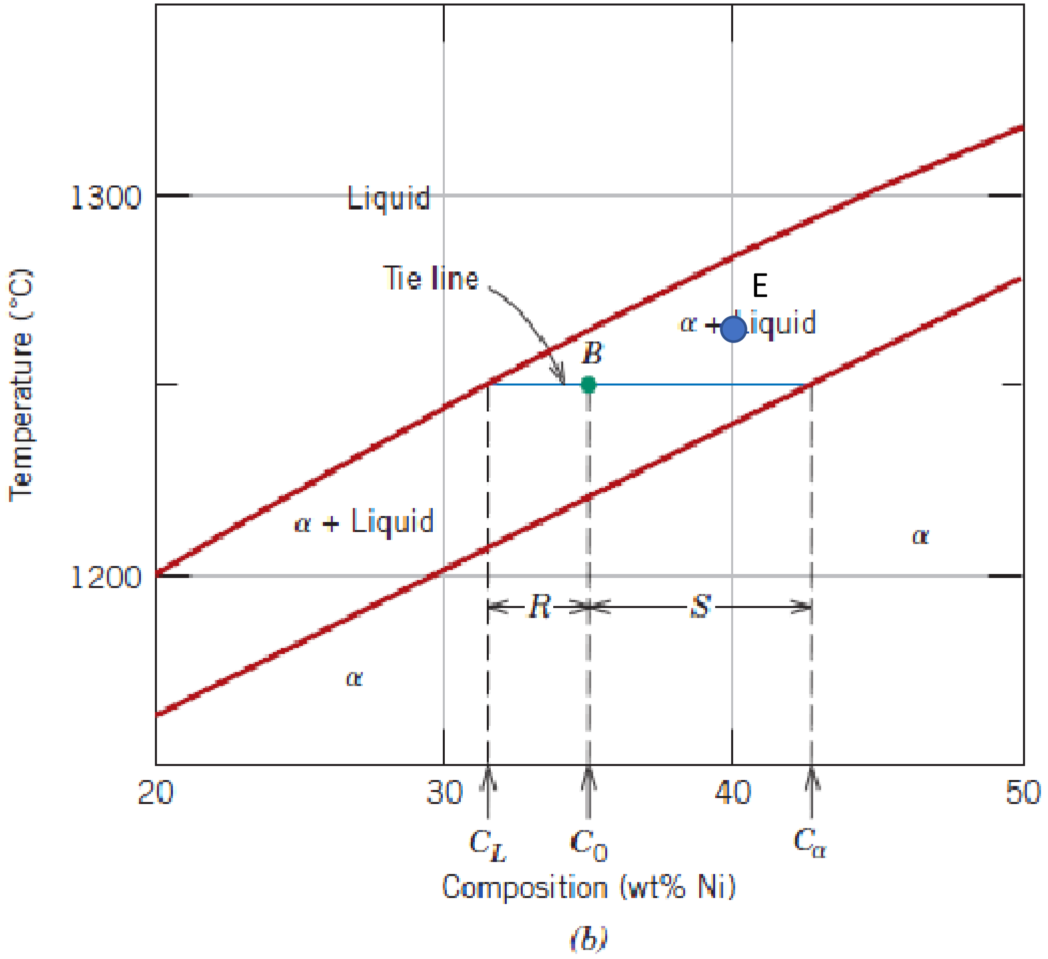


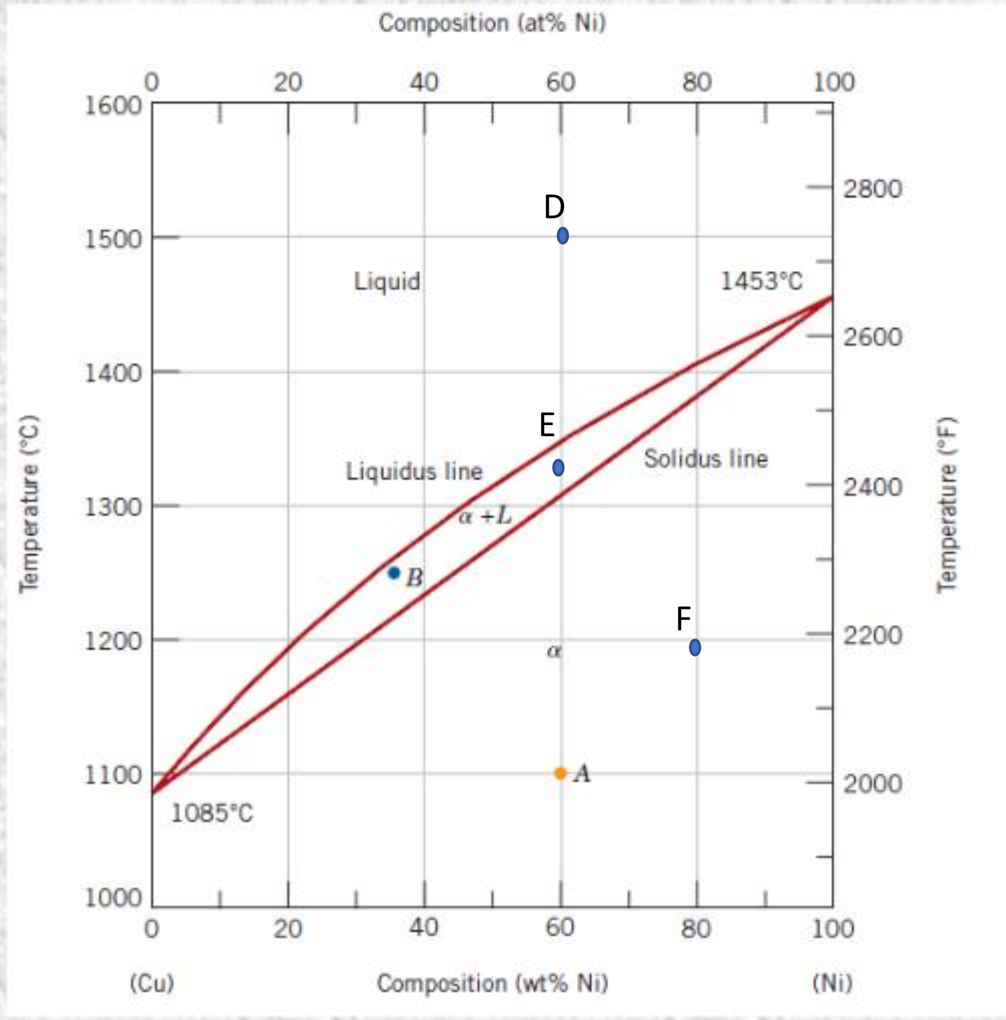
Point C:

Phases: 1 phase liquid
Composition: 20%Ni, 80% Cu
Relative amount: 100% liquid phase

Exercise 3: determine the phase present, their composition and relative amounts in the point D, E and F indicated in the phase diagram. Schematize the microstructure in each point.

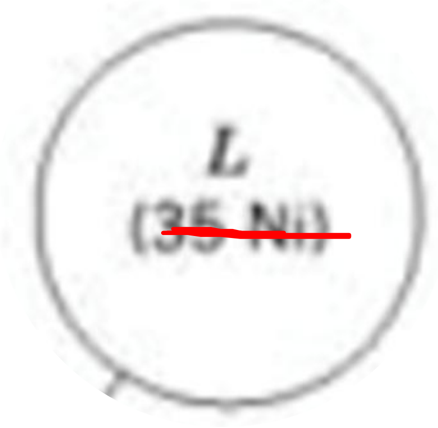


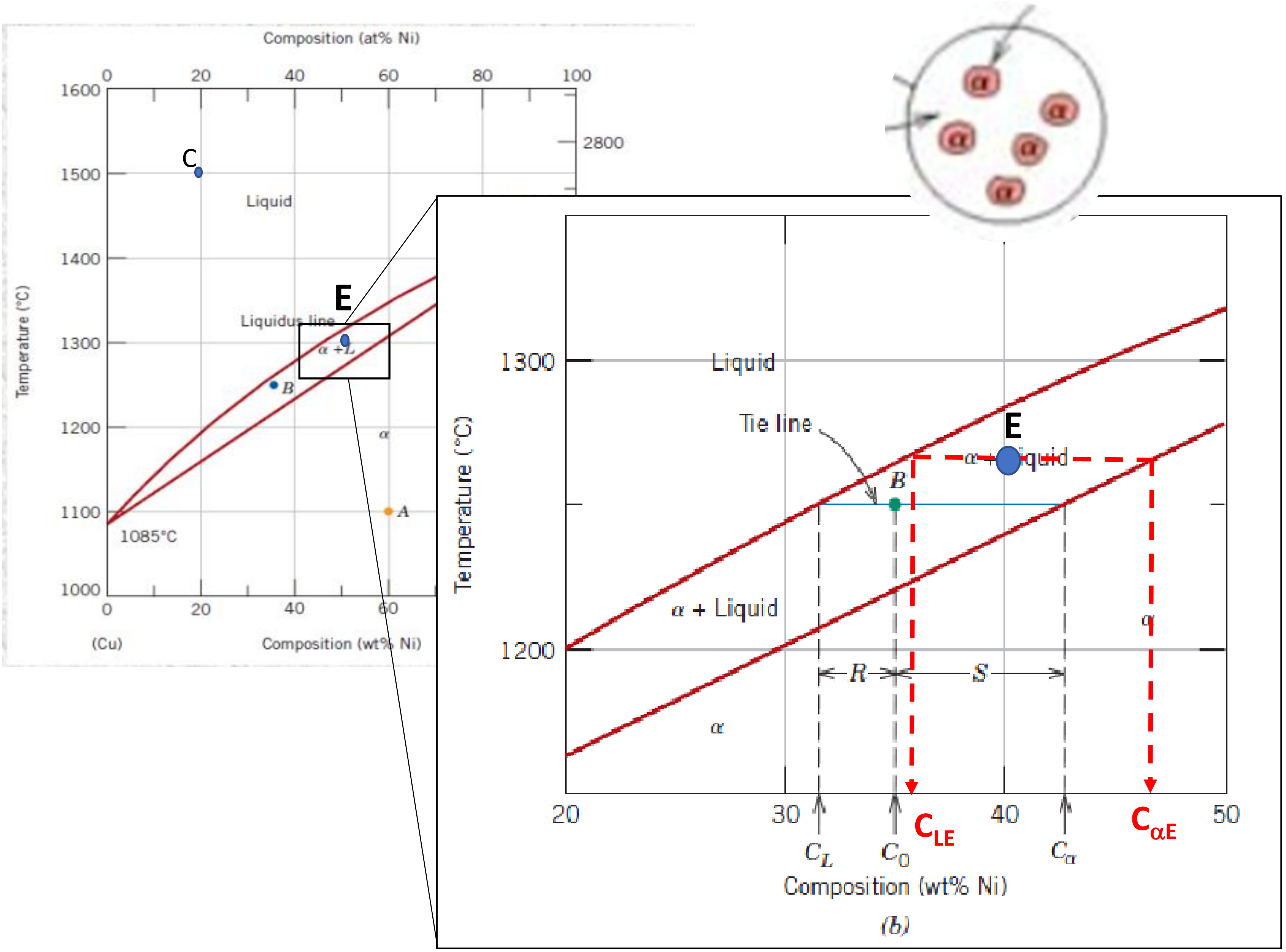




Point D:

Phases: 1 phase liquid
Composition: 60%Ni, 40% Cu
Relative amount: 100% liquid phase





Point E:

Phases: 2 phases =solid (α) +liquid (L)

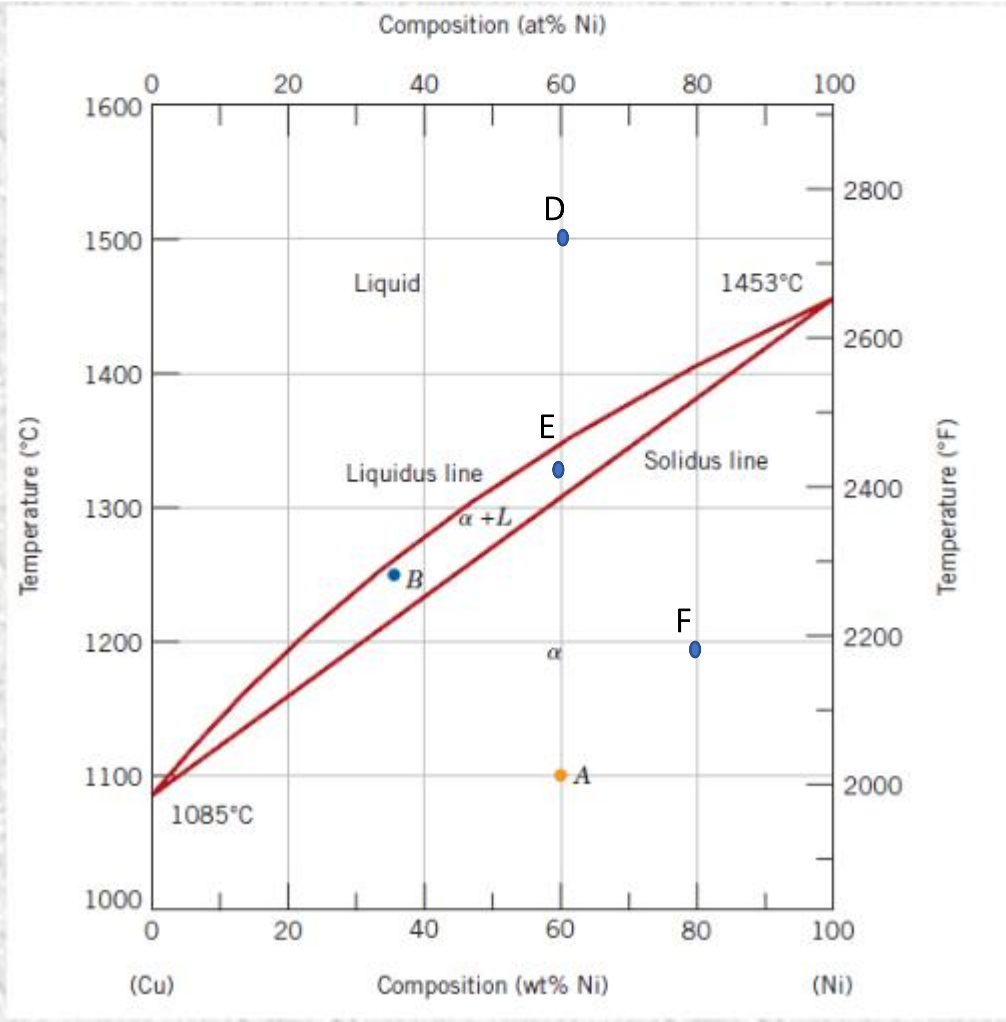
Composition: 2 phases region $\rightarrow$ horizontal rule

Composition of liquid (L):
C_{LE}= 36% Ni, 64%Cu

Composition of solid (α):
C αE= 46%Ni, 54% Cu

Relative amount : 2 phases region $\rightarrow$ lever rule

% α= (40-36)/(46-36)*100=40%
%L=(46-40)/(46-36)*100=60%

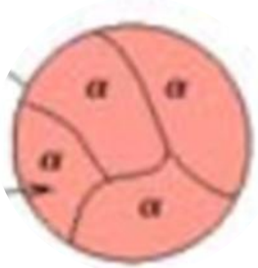


<u>Point F:</u>

Phases: 1 phase solid

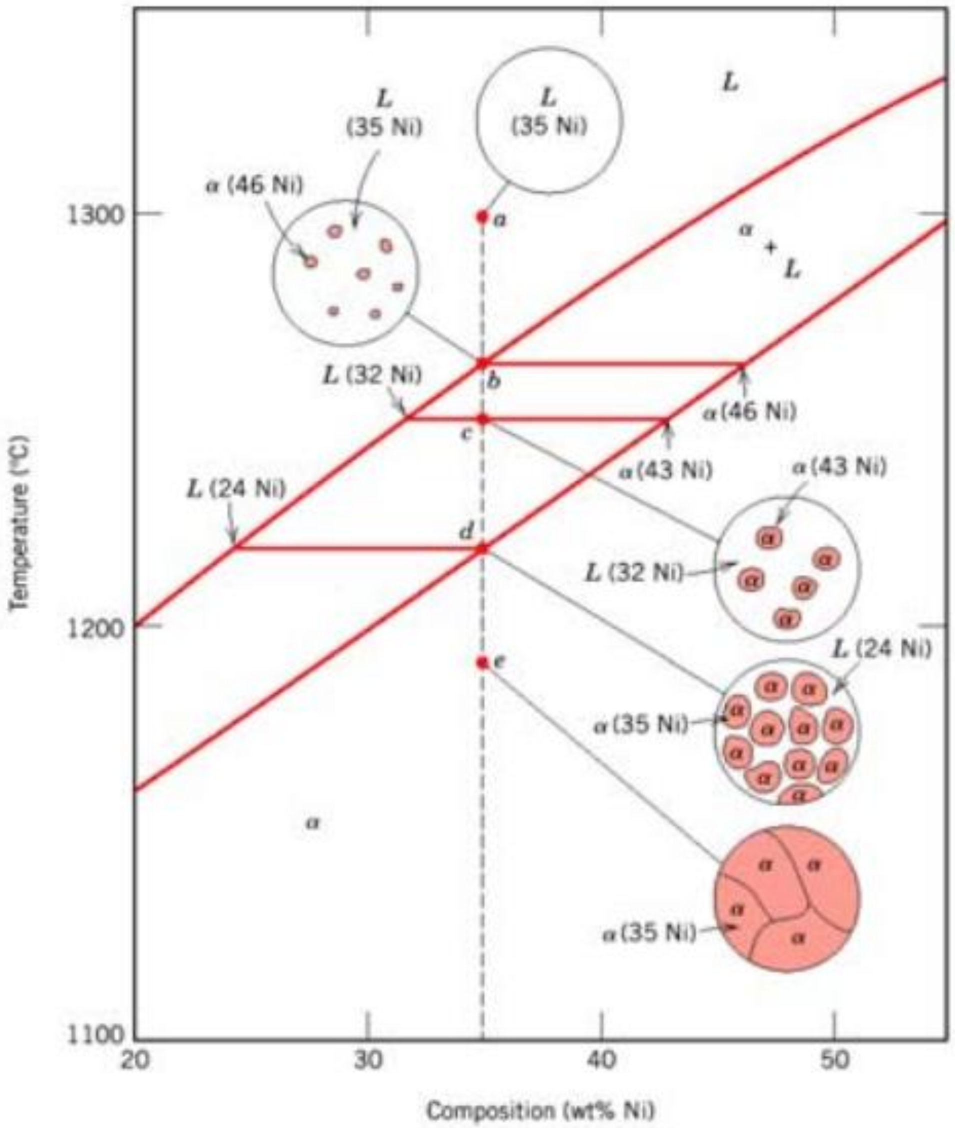
Composition: 80%Ni, 20% Cu

Relative amount: 100% solid phase



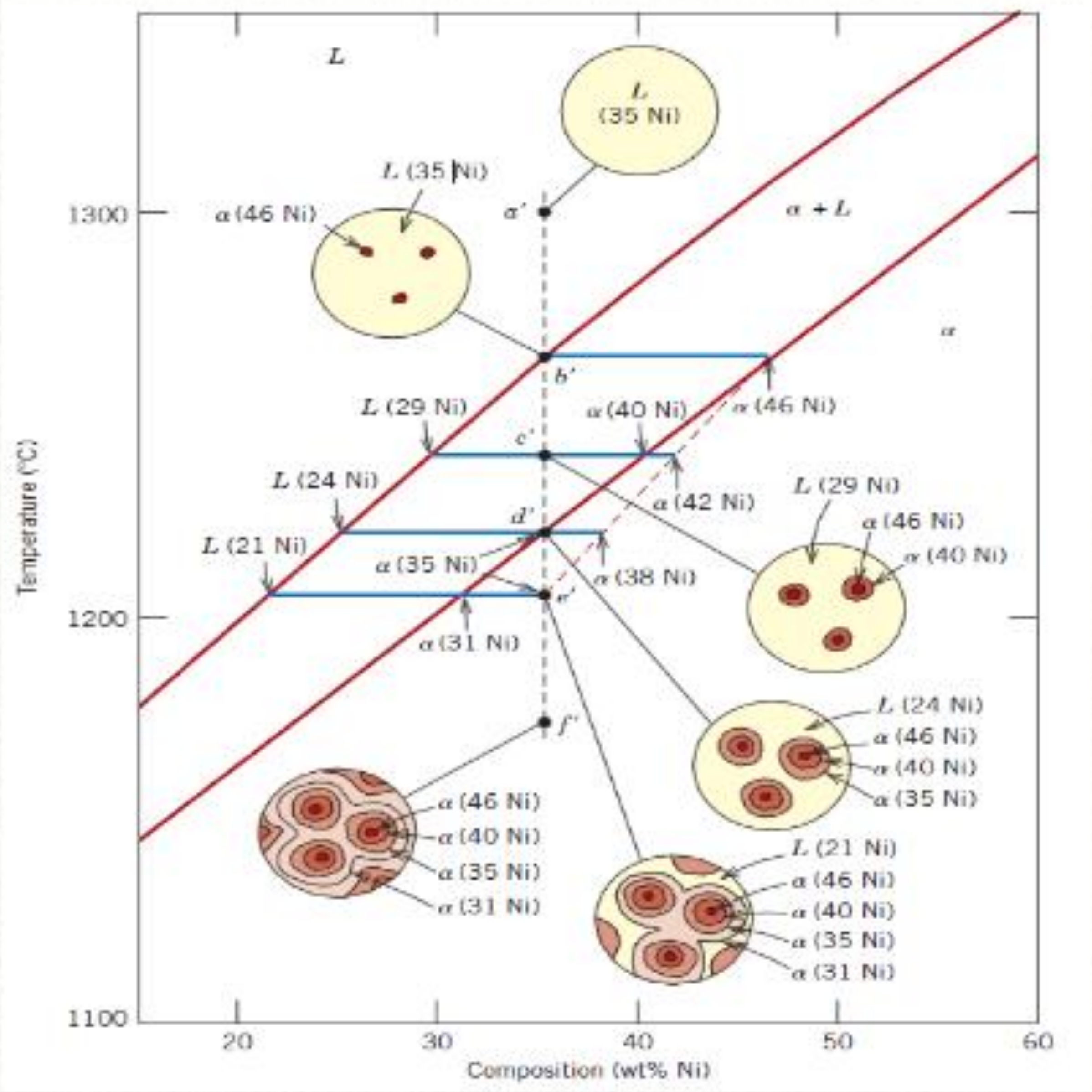
Development of microstructure in isomorphous alloys
Equilibrium (very slow) cooling

[redacted]

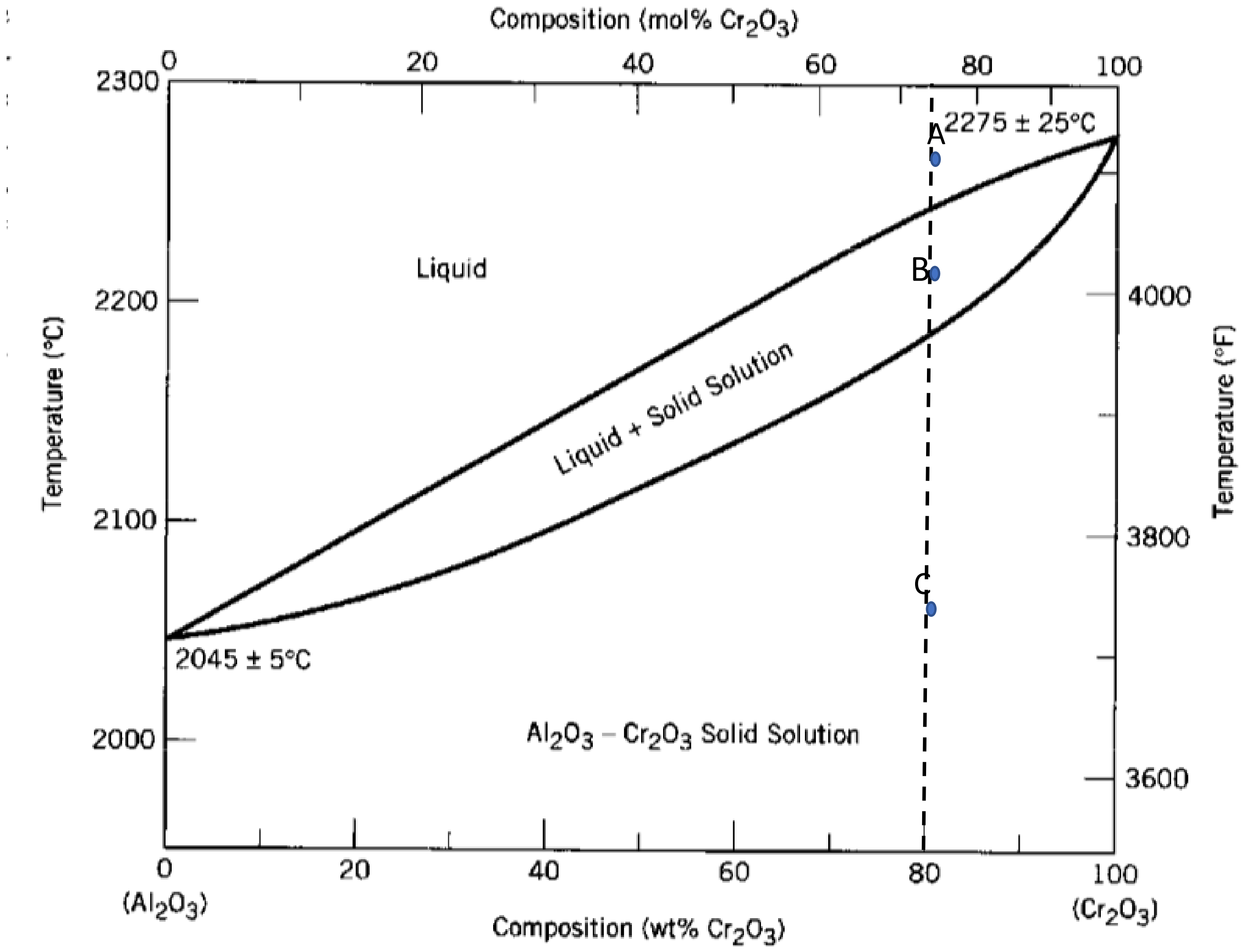


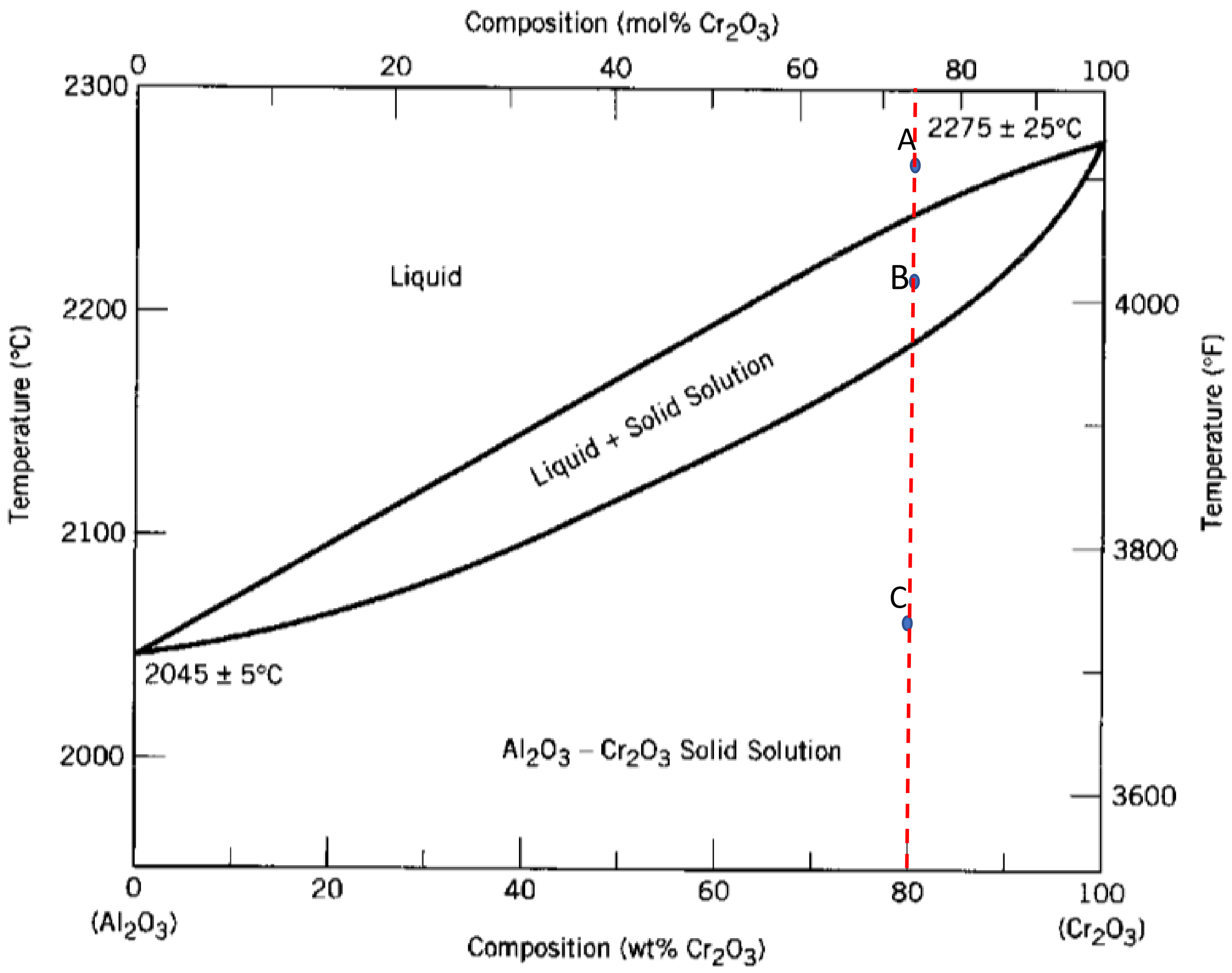
Non-Equilibrium Cooling

For Large cooling rate there is not enough time for diffusion!



Exercise 4: determine the phase present, their composition and relative amounts in the point A, B and C indicated in the phase diagram.



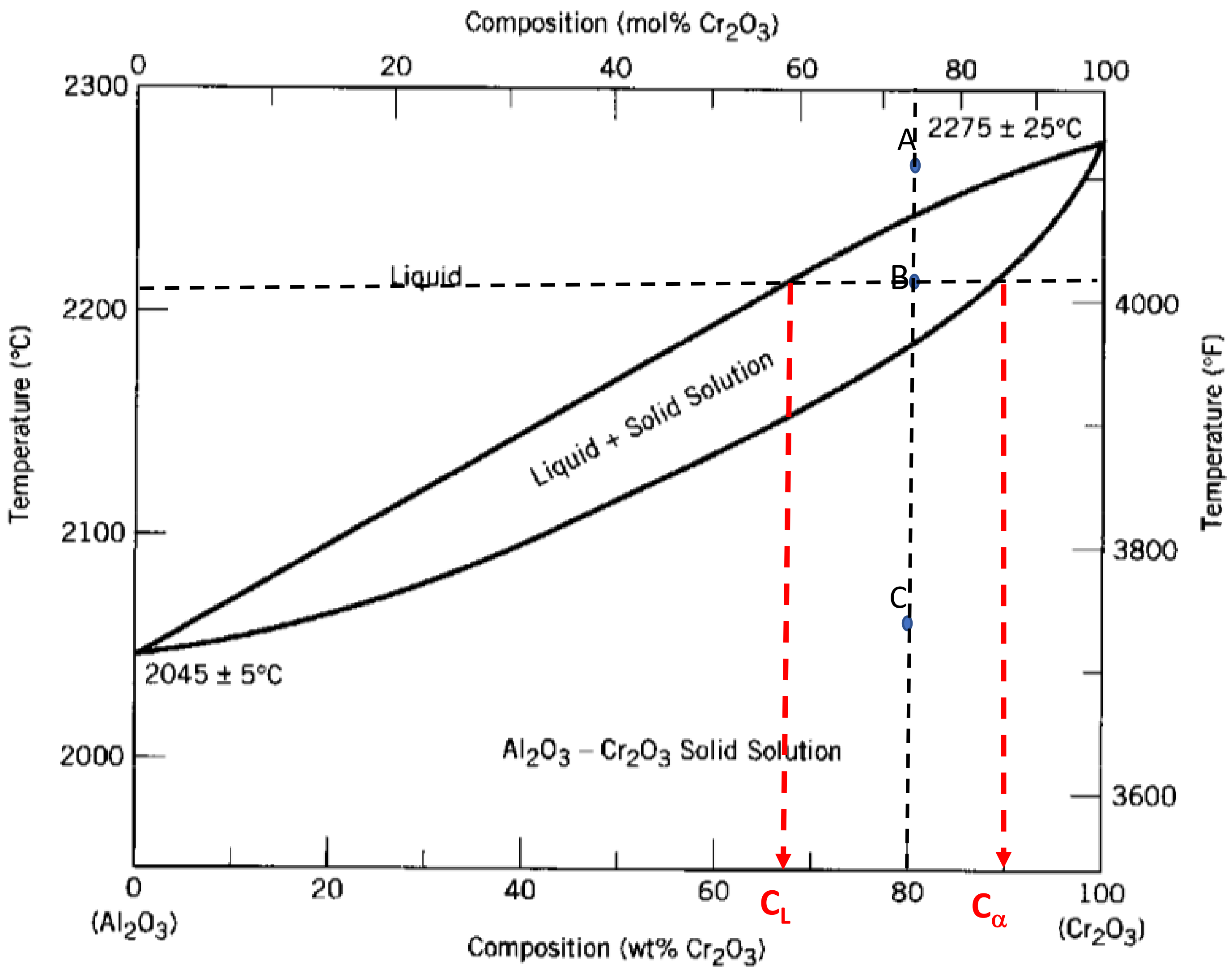


Point A:

Phases: 1 phase liquid

Composition: 80%Cr_2O_3, 20% Al_2O_3

Relative amount: 100% liquid phase



Point B:

Phases: 2 pahses solid (α) +liquid (L)

Composition: 2 phases region → horizontal rule

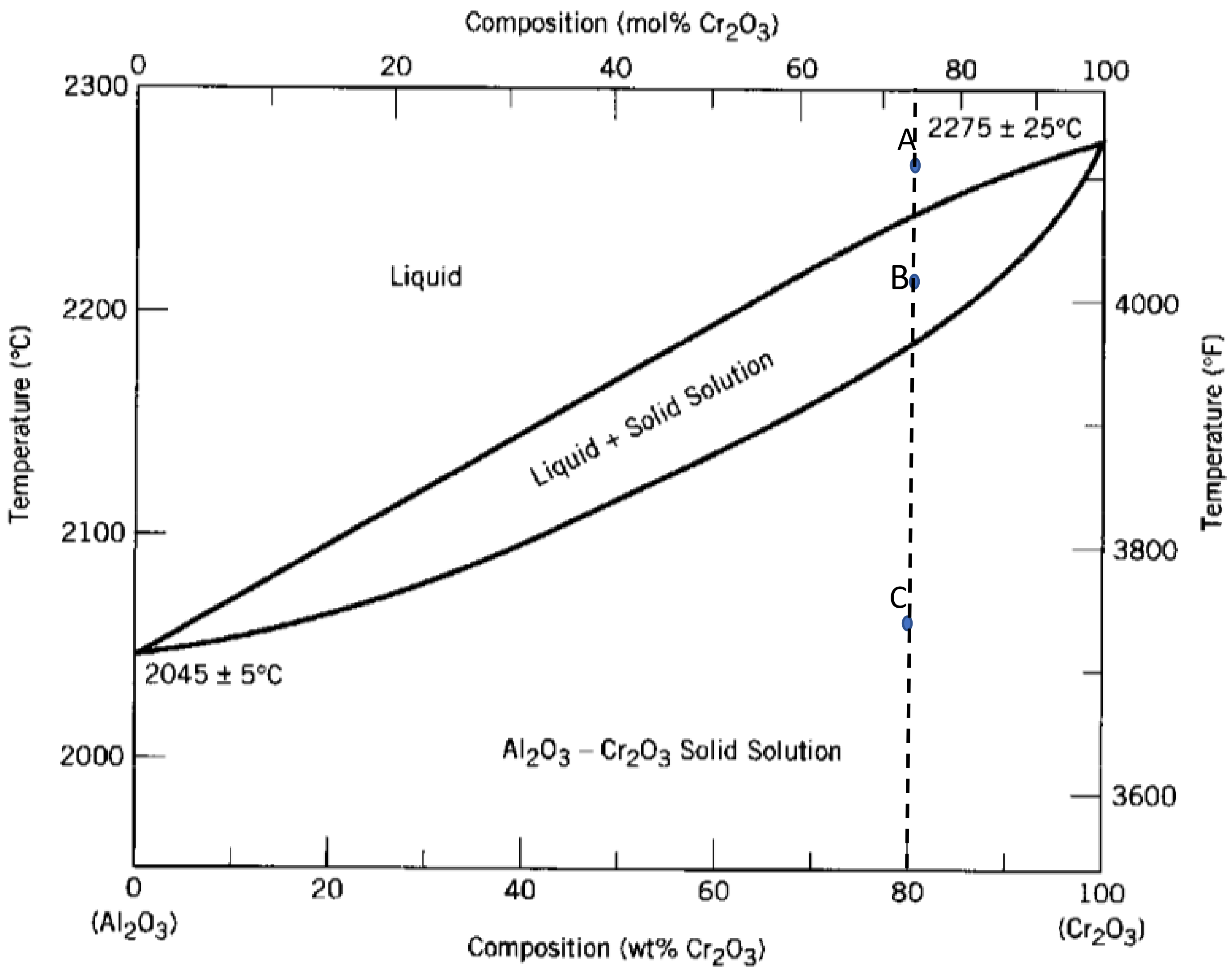
Composition of liquid (L): C_L= 68% Cr_2O_3, 32% Al_2O_3

Composition of solid (α): $C\alpha$= 90% Cr_2O_3, 10% Al_2O_3

Relative amount: 2 phases region → lever rule

% α= (80-68)/(90-68)*100=54.5%

%L=(90-80)/(90-68)*100=45.5%



Point C:

Phases: 1 phase solid

Composition: 80%Cr_2O_3, 20% Al_2O_3

Relative amount: 100% solid phase

Exercise 5: Draw a phase diagram (for generic components A and B), with **complete solubility in the liquid state** and **no solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

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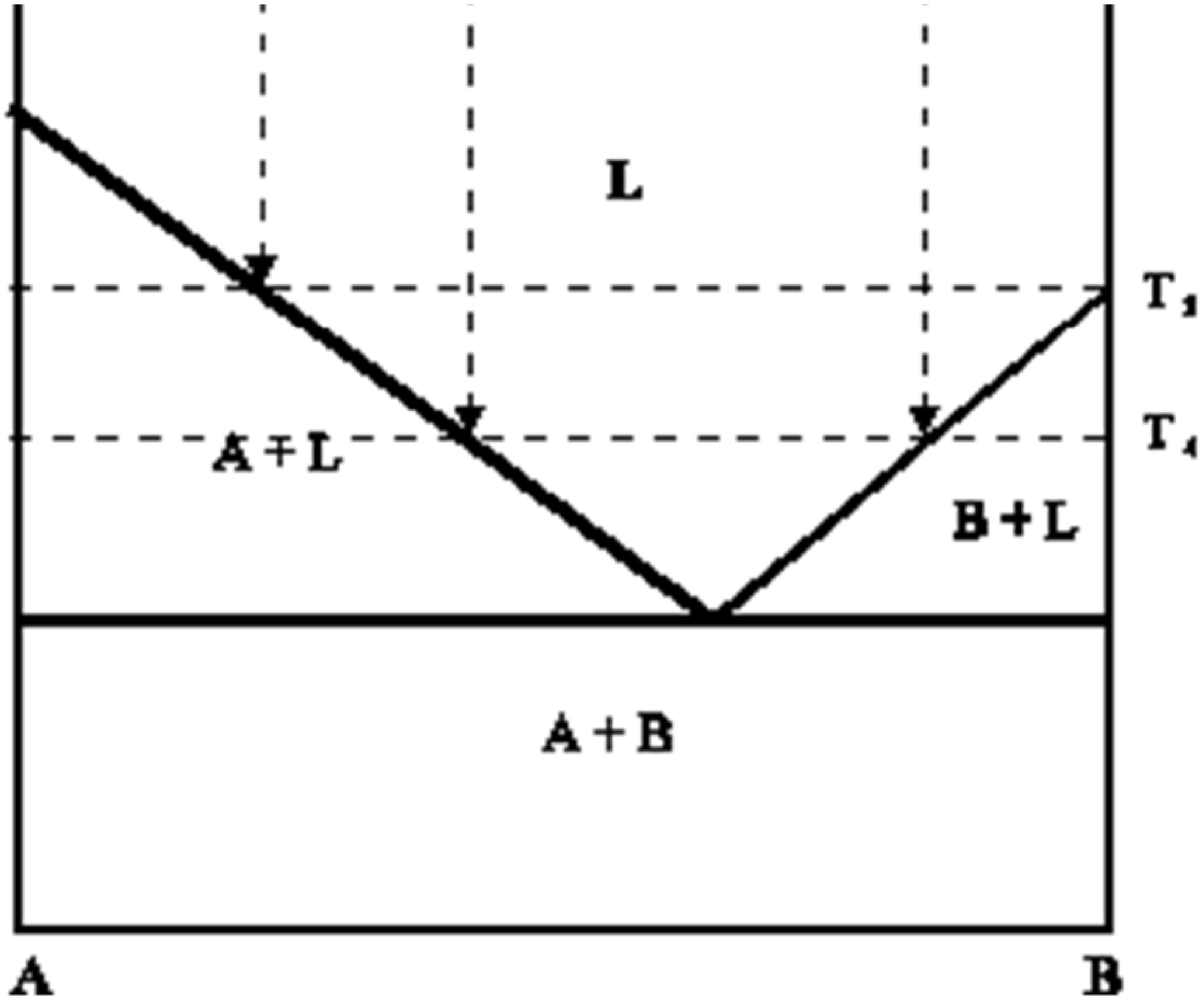
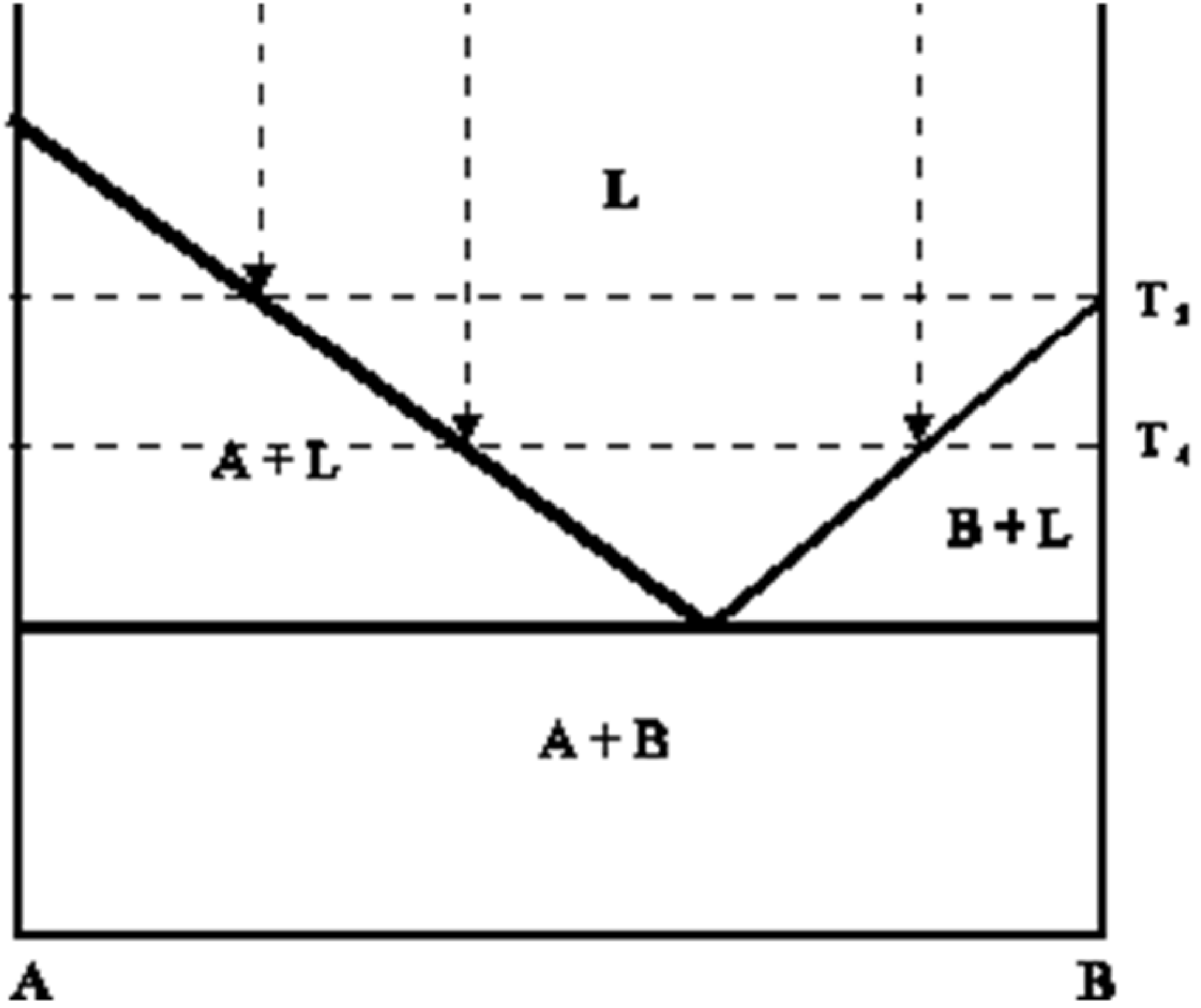


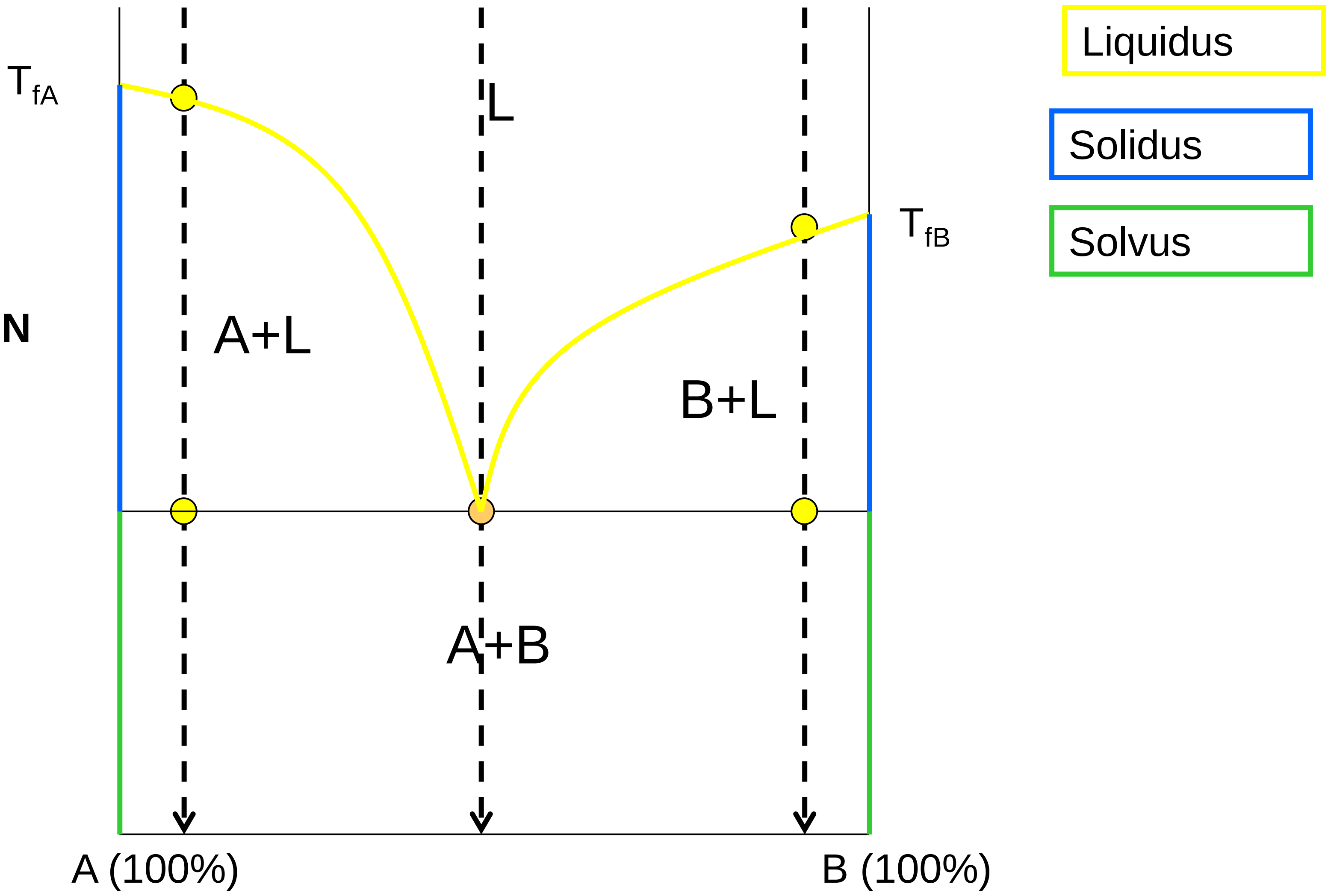
Complete solubility in the liquid state, no solubility in the solid state



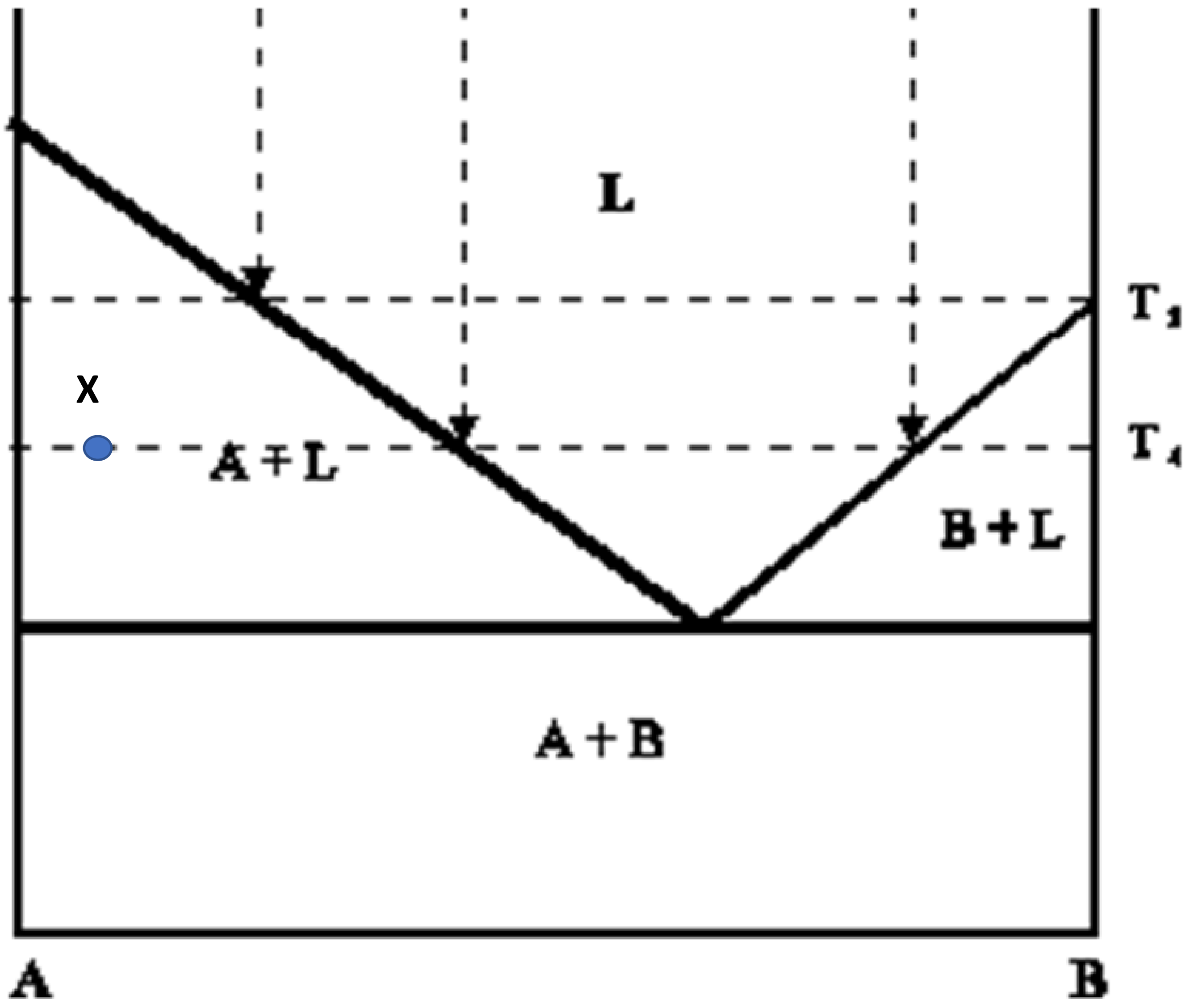
COMPLETE SOLUBILITY IN THE LIQUID STATE AND NO SOLUBILITY IN THE SOLID STATE, EUTECTIC

EUTECTIC TRANSFORMATION

L ↔ A+B



Exercise 6: determine the phases present, their compositions and relative amounts in the point X.



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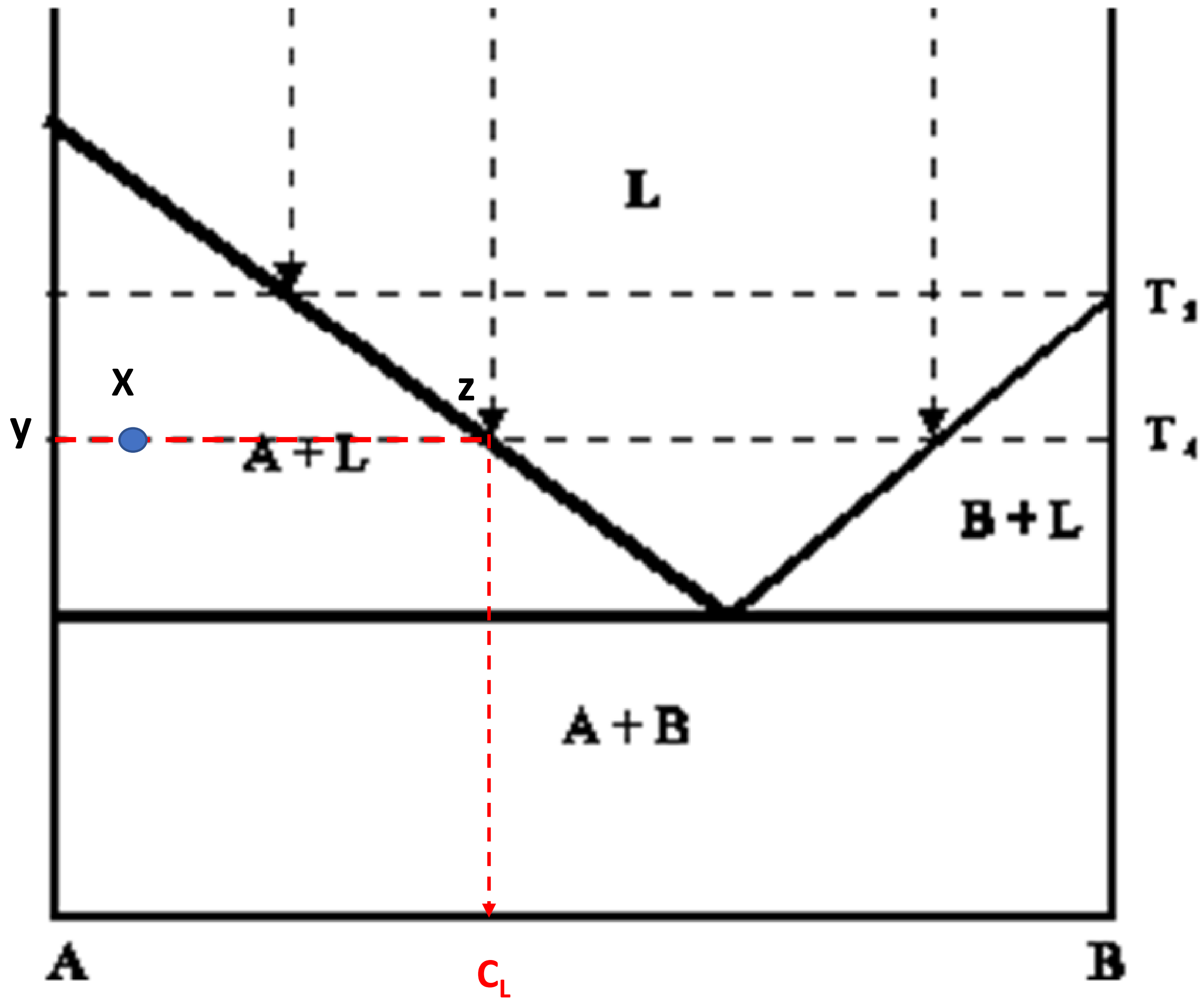


Point X

Phases: solid A+ Liquid L

Compositions: 2 phases region → horizontal rule

C_A: 100% A

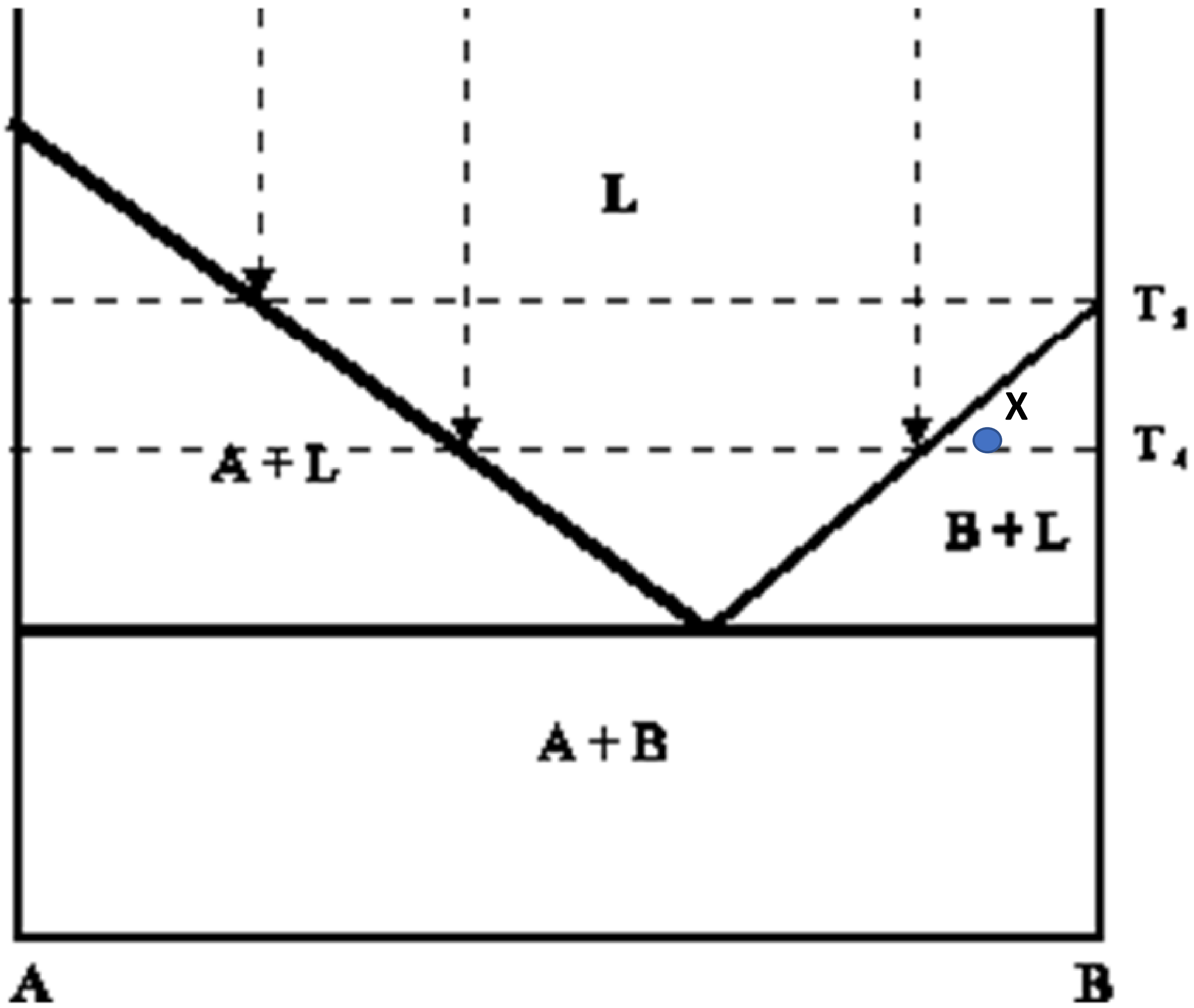
C_L= 45%B, 55%A

Relative amounts

%A= (Xz)/(yz)*100= (45-10)/(45-0)*100=78%

%L=(yX)/(yz)=(10-0)/(45-0)*100=22%

Exercise 7: determine the phases present, their compositions and relative amounts in the point X.



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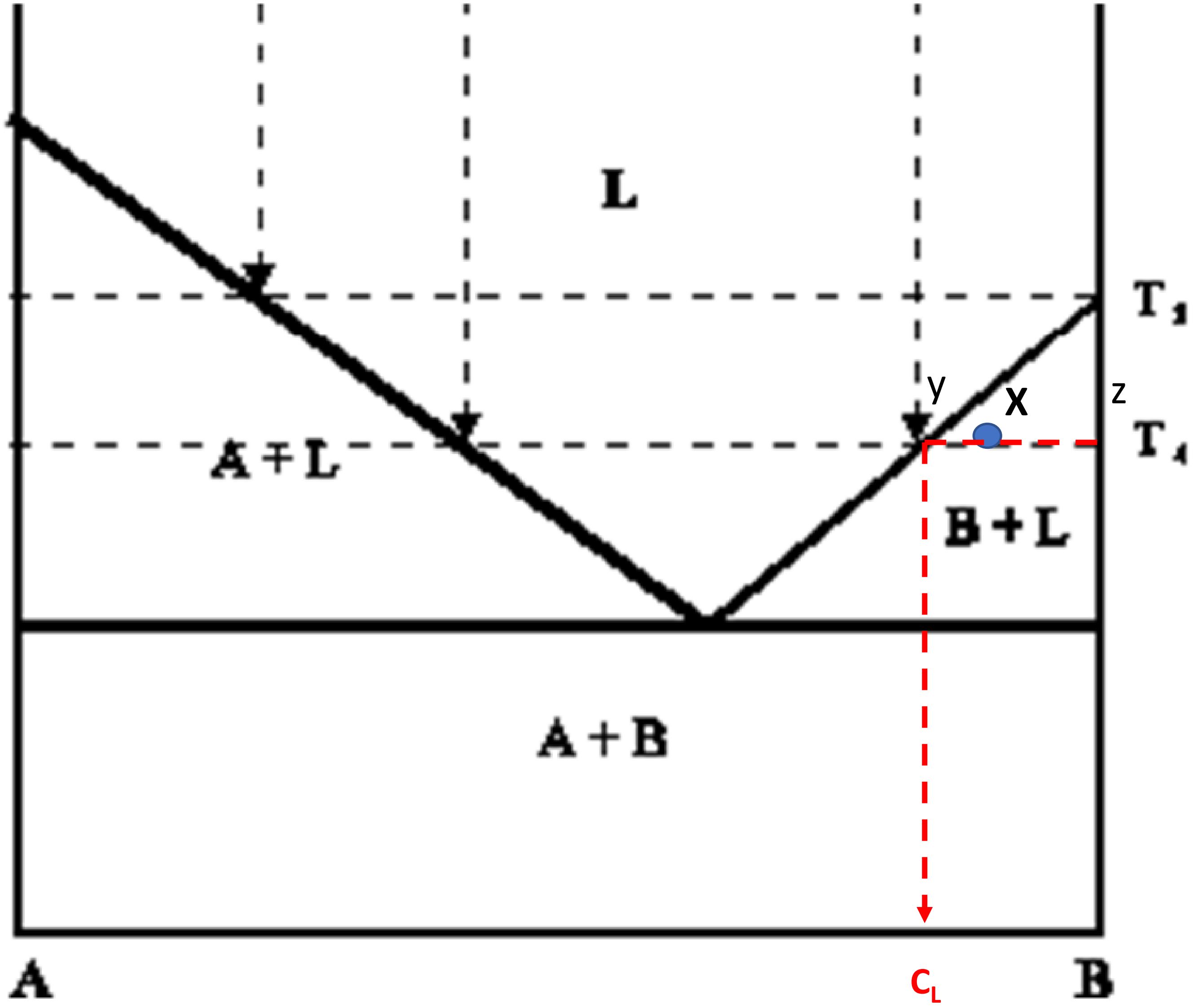


Point X

Phases: solid B+ Liquid L

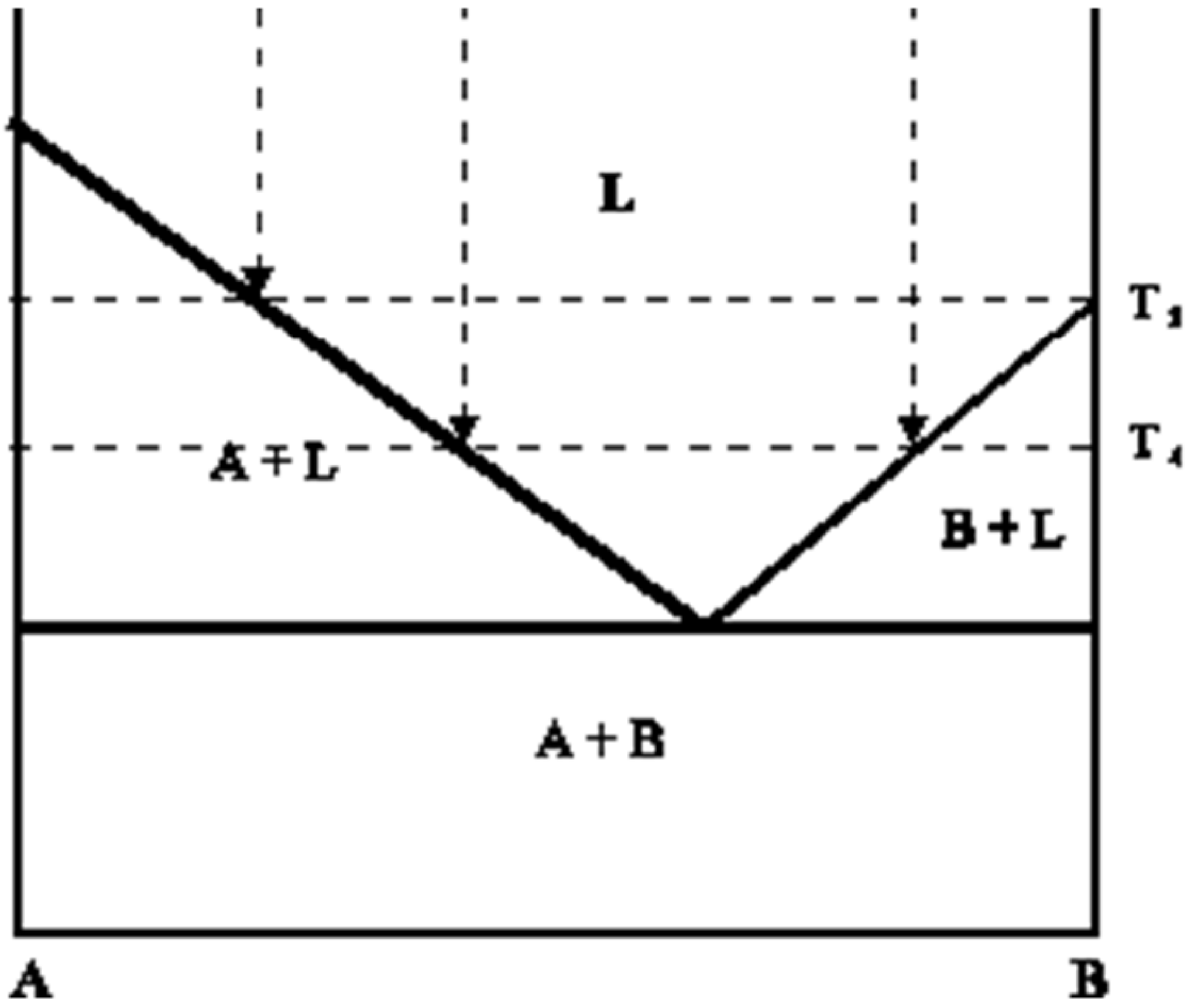
Compositions: 2 phases region → horizontal rule

C_B: 100% B
C_L= 80%B, 20%A

Relative amounts

%B = (Xy)/(yz)*100 = (85-80)/(100-80)*100 = =25%

%L = (Xz)/(yz) = (100-85)/(100-80)*100 = = 75%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T
T
A+L
B+L
A+B
A
B